Lab01 - Basic Electronics

MTRN3100 - UNSW School of Mechanical and Manufacturing Engineering

Task 0 - LED

Construct the circuit shown below. Use the provided wall socket for power.

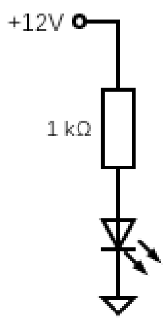

Figure 1: Circuit for task 0

Task 1 - Diode and Regulator

The datasheet for the voltage regulator can be found: here

Construct the circuit shown below and use a multimeter to verify the output voltages. **NOTE: Leave this circuit assembled on your breadboard, as you need the 5V supply for the following tasks**.

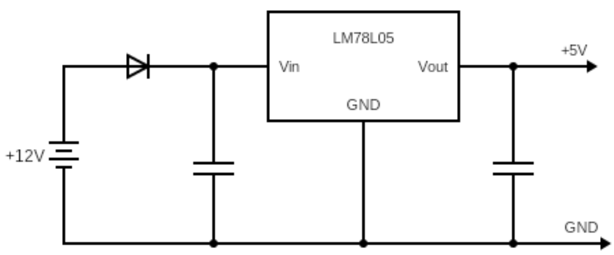

Figure 2: Circuit for task 1

Questions

- **Question 1:** What is the voltage drop off across the diode?

Task 2 - Button and Pulldown Resistor

You have been provided two buttons. Use the multimeter to identify and then select the nominally open button. Nominally open means there is no connection between the terminals unless the button is pressed.

Build the circuit below and use a multimeter to verify that the button works correctly. **NOTE: Leave this circuit assembled on your breadboard, as you need the Control signal for the next tasks**.

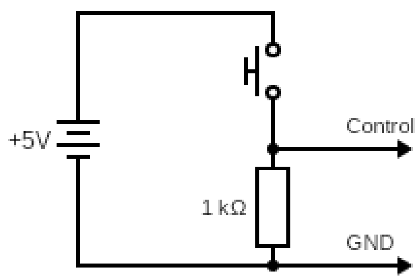

Figure 3: Circuit for task 2

Questions

- **Question 2:** If no resistor or wire is connected to the button (floating pin) what voltages do you observe?
- **Question 3:** Why is it better to use a resistor over a wire to pull a pin?

Task 3 - Motor Control

The datasheet for the MOSFET used in this task can be found: here. **NOTE: The standard symbol of a MOSFET contains the internal configuration; you do not need to wire this. Instead, look for the DRAIN, GATE, and SOURCE pins on the datasheet and use them to construct the circuit**.

Construct the circuit shown below and verify that the motor turns on when the button is pressed.

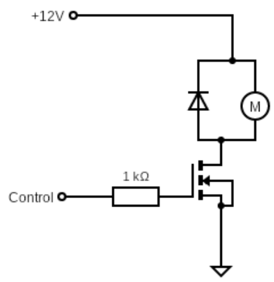

Figure 4: Circuit for task 5

Questions

- **Question 4:** What is the purpose of the protection diode in parallel with the motor? (Hint: If the system turns off, the DC motor will keep turning)
- **Question 5:** Why does the MOSFET have a large metal pad on the back?
- **Question 6:** What current does the motor draw with no load? (Hint: Place a 100ohm resistor in series with the motor and measure the voltage across it.)
- **Question 7:** Explain why the motor is powered from +12V and not from Control. (Hint: Control could come from a microcontroller)